MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6397

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Tetrode, rf beam power output

Outline	---	3-8 (EIA)
Base	---	E8-10
Envelope	---	T3
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	(B) 7	8
Element	---	-f	nc	a	nc	g3, fct +f, par	g2	+f, ser -f, par	g1

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ik	(B) Ic1(note 1)	Pp	Pg2	(B) Alt
Unit:	Vdc	Vdc	Vdc	Vdc	mAdc	μAdc	W	W	ft
Maximum:	1.4 or 2.8	135	---	135	14	375	1.5	0.6	(See note 9)
Minimum:	1.1 or 2.2	---	-100	---	---	---	---	---	---
TEST CONDITIONS:	1.25	125	-7.5	125	---	---	---	---	---

GENERAL:

Qualification - Required

(B) denotes changes

6397

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1121	Total grid current	See note 2	0.65	II	Ic1	0	-0.5	µAdc
1256	Electrode current (anode)		0.65	II	Ib	4.5	9.5	mAdc
1256	Electrode current (screen)		0.65	II	Ic2	0.6	1.6	mAdc
1306	Transconductance (1)	See note 2	0.65	II	Sm	1,500	2,400	µmhos
(B) 1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
(B) 1211	Insulation of electrodes	Ef = 2.5 V; Ec1 = -100 Vdc	4.0	S3	R	100	---	Meg
(B) 1301	Filament current		---	---	If	115	135	mAdc
(B) 1306	Transconductance (2)	Ef = 1.1 Vdc	---	---	Sm	1,300	---	µmhos
(B) ---	Doubler operation (1)	Eb = Ec2 = 120 Vdc; csig = 80 v; Rg1 = 0.22 Meg; Fin = 125 MHz; Fout = 250 MHz; t = 300 sec (see notes 3, 4, and 5)	---	---	Po Ic1	90 280	--- ---	mW µAdc
(B) ---	Doubler operation (2)	Ef = 1.1 Vdc; Eb = Ec2 = 81 Vdc; esig = 40 v; Rg1 = 0.22 Meg; t = 15 minutes (see note 4)	---	---	Po	20	---	mW
(B) ---	Doubler operation (3)	Eb = Ec2 = 120 Vdc; esig = 65 v; Rg1 = 0.27 Meg; Fin = 60 MHz; Fout = 120 MHz (see notes 6 and 7)	---	---	Po	80	---	mW
(B) 1331	Direct-interelectrode capacitance	With 0.405-inch diameter shield	6.5	Code E	Cg1p Cin Cout	--- 2.3 2.5	0.055 3.2 3.5	pF pF pF
(B) 1031	Low-frequency vibration	Rl = 2,000 ohms	6.5	Code G	Ep	---	200	mVac
1116	Lead fatigue		4.0	S3	---	---	---	---
(B) 1041	Shock	540 G	15	S4	---	---	---	---
(B) 1031	Vibration fatigue		6.5	Note 10	---	---	---	---
---	Post-shock and vibration-fatigue test end points	Low-frequency vibration	---	---	Ep	---	210	mVac
		Transconductance (1)	---	---	Sm	1,450	2,450	µmhos
(B) 1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A; self oscillator; F = 125 MHz; Rg1 = 0.15 Meg; Ic1 = 350 μAdc (see note 8)	---	---	---	---	---	---
---	Life-test end points (150 hours)	Doubler operation (1) Doubler operation (2)	--- ---	--- ---	Po Po	60 15	--- ---	mW mW
(B)	Periodic-check test:							
1202	Filament burnout		---	---	---	---	---	---

NOTES:

1. The maximum dc grid No. 1 current rating shall apply to frequency doubler applications.
2. This test to be performed at the conclusion of the holding period.
3. The circuit shall be adjusted for maximum power output to the load with Ib within the ranges of 5.5 mAdc minimum to 10.5 mAdc maximum. Po shall be the useful power output at the end of the specified period.
4. Test in 125 to 250 MHz doubler circuit per Drawing 218-JAN, with values as specified in test conditions. Po shall be useful power output.
5. Sample size shall be as specified by the quality conformance inspection, part 2 test. At least five samples of each lot shall be subjected to doubler operation (1) test for a period of 30 minutes. More than one reject out of the five tubes shall constitute a lot rejection. Total rejects, from the 5 minute and the 30-minute tested samples, in excess of quantities permitted by the quality conformance inspection, part 2 test, shall constitute a lot rejection. Not less than five tubes shall be tested when reduced sampling is in effect.
6. The circuit shall be adjusted for maximum power output to the load with Ib within the range of 5.0 mAdc minimum to 9.0 mAdc maximum. Po shall be the useful power output.
7. Test in 60 to 120 MHz doubler circuit per Drawing 218-JAN. It is recommended that a reproduction of the circuit of Radio Set AN/URC-4 be used as the test equipment.
8. In a self oscillator circuit, the load is adjusted so that the average Ib = 7 mAdc.
9. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
10. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 80
DSA - ES

User activities:
Army - MU, WC
Navy - AS, OS, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-78)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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